

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Week No.: 2

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Date: 30-08-2026

1. TITLE

Design Webpage Using HTML5 & CSS3

2. AIM

To construct a styled webpage using HTML5 semantic elements and CSS3 layout styles.

3. OBJECTIVE

The objectives of this experiment are:

- Implement CSS3 rules and flexbox layouts.
- Style header and cards.

4. THEORY

CSS3 provides custom background gradients, flexbox alignment, and box shadow effects.

5. ALGORITHM / PROCEDURE

1. Create index.html and style.css.
2. Apply linear gradient background.
3. Build card layout with flexbox.
4. Verify in browser.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Web Development Portal</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
  <h1>Web Development Portal</h1>
  <p>Full Stack Lab Practice</p>
</header>

<nav>
```

```
<a href="#">Home</a>
<a href="#">Courses</a>
<a href="#">Projects</a>
<a href="#">Contact</a>
</nav>

<section class="container">

  <div class="card">
    <h2>HTML5 Structure</h2>
    <p>Provides semantic structure and tags for web pages.</p>
    <button>Explore</button>
  </div>

  <div class="card">
    <h2>CSS3 Styling</h2>
    <p>Adds vibrant colors, flexbox grid, and smooth hover effects.</p>
    <button>Explore</button>
  </div>

  <div class="card">
    <h2>JavaScript Logic</h2>
    <p>Enables dynamic interactivity and DOM manipulation.</p>
    <button>Explore</button>
  </div>

</section>

<footer>
  <p>© 2026 B.Harsha Vardhan | Full Stack Lab</p>
</footer>

</body>
</html>
```

style.css

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: #eef2f5;
}

header {
  background: linear-gradient(135deg, #1e3c72, #2a5298);
  color: white;
  text-align: center;
  padding: 22px;
}

nav {
  background: #111;
  text-align: center;
  padding: 12px;
}

nav a {
  color: white;
  text-decoration: none;
```

```
    margin: 15px;
    font-size: 16px;
}

nav a:hover {
    color: #4facfe;
}

.container {
    display: flex;
    justify-content: center;
    gap: 20px;
    margin: 30px 20px;
}

.card {
    background: white;
    width: 240px;
    padding: 20px;
    border-radius: 12px;
    text-align: center;
    box-shadow: 0 4px 10px rgba(0,0,0,0.1);
}

button {
    background: #1e3c72;
    color: white;
    border: none;
    padding: 8px 16px;
    border-radius: 6px;
    cursor: pointer;
}

button:hover {
    background: #2a5298;
}

footer {
    background: #111;
    color: white;
    text-align: center;
    padding: 12px;
}
```

7. CODE EXPLANATION

Code / Syntax	Explanation
display: flex;	Flexbox container layout
background: linear-gradient(...)	Gradient background style

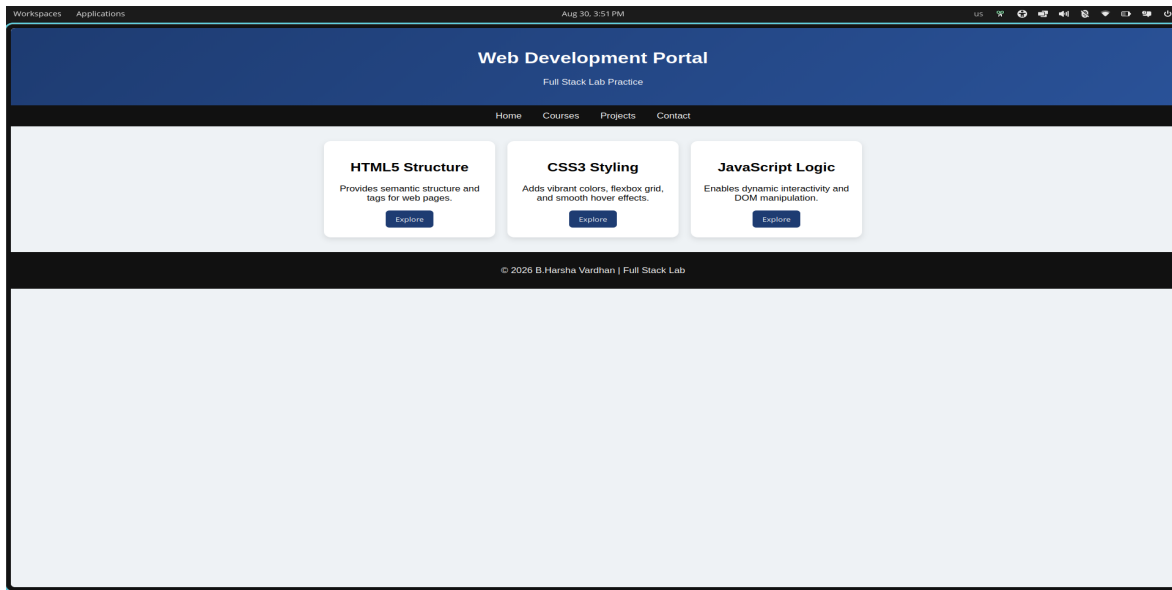
8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Load	Render styled layout	Render styled layout	Pass

9. OUTPUT

Output 1: Execution Screenshot



10. RESULT

Thus, the experiment for Week 2 (Design Webpage Using HTML5 & CSS3) was successfully executed and verified.